

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0684

Swansea 138 kV
September 30, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0684. The Interconnection Customer proposes a 407.3 MW solar photovoltaic facility interconnecting at the Swansea 138 kV substation in the Delaware area. The request was received on October 04, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 229 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$84,484,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Sheffield Wind - Swansea 138kV L
Contingency: Tombstone - Bixby Hill 500kV
Loading: 242.6% (Rating: 204 MVA)
DFAX: 68.4%
Pre-project loading: 105.9%
2. Facility: Swansea - Clay 138kV Line
Contingency: Swansea - Sheffield Wind 138kV
Loading: 230.8% (Rating: 161 MVA)
DFAX: 100.0%
Pre-project loading: 22.2%
3. Facility: Sheffield Wind - Dolphus M Grainger 138kV
Contingency: Tombstone - Bixby Hill 500kV Line
Loading: 208.8% (Rating: 222 MVA)
DFAX: 68.4%
Pre-project loading: 83.4%
4. Facility: Dolphus M Gringer - Aztec 138kV
Contingency: Tombstone - Bixby Hill 500kV
Loading: 198.9% (Rating: 229 MVA)
DFAX: 68.4%
Pre-project loading: 76.9%
5. Facility: Clay - S. Macon 138kV line
Contingency: Sheffield Wind - Swansea 138kV
Loading: 197.5% (Rating: 184 MVA)
DFAX: 100.0%
Pre-project loading: 24.2%

6. Facility: Deer Vly - Chimayo 138kV
Contingency: Aztec 345 kV/138 kV Transformer
Loading: 177.4% (Rating: 149 MVA)
DFAX: 56.9%
Pre-project loading: 22.2%
7. Facility: Solway - S Macon 138kV
Contingency: Sheffield Wind Substation - Swansea Substation 138kV
Loading: 169.8% (Rating: 204 MVA)
DFAX: 100.0%
Pre-project loading: 30.3%
8. Facility: Leander - Solway 138kV line
Contingency: Sheffield Wind - Swasea 138kV
Loading: 159.1% (Rating: 211 MVA)
DFAX: 100.0%
Pre-project loading: 34.2%
9. Facility: Leander - Petersburg 138kV
Contingency: Sheffield Wind - Swansea 138kV
Loading: 150.1% (Rating: 215 MVA)
DFAX: 100.0%
Pre-project loading: 39.3%
10. Facility: Port Hudson - Chimayo Substation 138kV
Contingency: Aztec 345 kV/138 kV Transformer Bank
Loading: 149.8% (Rating: 184 MVA)
DFAX: 56.9%
Pre-project loading: 23.9%
11. Facility: Port Hudson - Aztec 138kV
Contingency: Aztec 345 kV/138 kV Autotransformer
Loading: 145.8% (Rating: 206 MVA)
DFAX: 56.9%
Pre-project loading: 33.5%
12. Facility: Glasgow - Dell Crk 138kV L
Contingency: Hope Crk - Star Peak 345kV
Loading: 145.0% (Rating: 180 MVA)
DFAX: 5.1%
Pre-project loading: 133.4%
13. Facility: Mill Cteek - Blooming Prairie 138kV
Contingency: Hope Crk - Star Peak 345kV
Loading: 140.6% (Rating: 214 MVA)
DFAX: 5.1%
Pre-project loading: 130.9%
14. Facility: Glagow - Blooming Prairie 138kV
Contingency: Star Peak - Hope Creek 345kV
Loading: 139.8% (Rating: 209 MVA)
DFAX: 5.1%
Pre-project loading: 129.8%
15. Facility: Neevin - Petersburg 138kV
Contingency: Swansea - Sheffield Wind 138kV Line
Loading: 127.6% (Rating: 243 MVA)
DFAX: 100.0%
Pre-project loading: 40.3%
16. Facility: Deer Valley - Newlinville 138kV L
Contingency: Aztec 345/138 kV TX

Loading: 116.3% (Rating: 201 MVA)
 DFAX: 56.9%
 Pre-project loading: 1.2%

17. Facility: Otay Tap - Denham Springs 138kV L
 Contingency: Hope Crk - Star Peak 345kV
 Loading: 114.6% (Rating: 235 MVA)
 DFAX: 7.0%
 Pre-project loading: 102.4%

18. Facility: Otay Tap Substation - Shred 138kV Line
 Contingency: Hope Creek - Star Pek 345kV Line
 Loading: 108.4% (Rating: 220 MVA)
 DFAX: 7.0%
 Pre-project loading: 95.5%

19. Facility: Shred - Seminole 138kV line
 Contingency: Star Peak - Hope Creek 345kV
 Loading: 104.6% (Rating: 200 MVA)
 DFAX: 7.0%
 Pre-project loading: 90.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sheffield Wind - Swansea 138kV line	Rebuild line with higher-capacity conductor	\$2,215,000	100% (\$2,215,000)
Clay - Swansea 138kV	Rebuild line with higher-capacity conductor	\$2,218,000	100% (\$2,218,000)
Sheffield Wind - Dolphus M Grainger 138kV	Rebuild line with higher-capacity conductor	\$4,360,000	100% (\$4,360,000)
Dolphus M Grainger - Aztec 138kV	Rebuild line with higher-capacity conductor	\$1,623,000	100% (\$1,623,000)
Clay - S. Macon 138kV line	Rebuild line with higher-capacity conductor	\$5,621,000	100% (\$5,621,000)
Deer Vly - Chimayo 138kV	Rebuild line with higher-capacity conductor	\$4,104,000	100% (\$4,104,000)
S. Macon - Solway 138kV	Rebuild line with higher-capacity conductor	\$5,702,000	100% (\$5,702,000)
Leander - Solway 138kV line	Rebuild line with higher-capacity conductor	\$3,182,000	100% (\$3,182,000)
Leander - Petersburg 138kV	Rebuild line with higher-capacity conductor	\$9,600,000	100% (\$9,600,000)
Port Hudson - Chimayo Substation 138kV	Rebuild line with higher-capacity conductor	\$2,824,000	100% (\$2,824,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Aztec - Port Hudson 138kV line	Rebuild line with higher-capacity conductor	\$2,049,000	100% (\$2,049,000)
Glasgow - Dell Crk 138kV L	Rebuild line with higher-capacity conductor	\$2,963,000	100% (\$2,963,000)
Mill Cteek - Blooming Prairie 138kV	Rebuild line with higher-capacity conductor	\$3,392,000	100% (\$3,392,000)
Glagow - Blooming Prairie 138kV	Rebuild line with higher-capacity conductor	\$19,370,000	100% (\$19,370,000)
Neevin - Petersbug 138kV	Rebuild line with higher-capacity conductor	\$4,304,000	100% (\$4,304,000)
Deer Valley - Newlinville 138kV L	Rebuild line with higher-capacity conductor	\$2,543,000	100% (\$2,543,000)
Otay Tap - Denham Springs 138kV L	Rebuild line with higher-capacity conductor	\$6,541,000	100% (\$6,541,000)
Shred - Otay Ap 138kV line	Reconductor line	\$847,000	100% (\$847,000)
Shred - Seminole 138kV line	Reconductor line	\$1,026,000	100% (\$1,026,000)
	Total allocated to project		\$84,484,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.